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Low-Cost Storm and Lightning Warning Box with Two-Tier Cloud Check, Offline Fallback, and Bangla Voice Alert for Rural Farmers

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
BANGLADESH ARMY INTERNATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (BAIUST)

DECLARATION & AI-USE DISCLOSURE

We declare that this CSE 301 Integrated Design Project proposal, "Low-Cost Storm and Lightning Warning Box with Two-Tier Cloud Check, Offline Fallback, and Bangla Voice Alert for Rural Farmers", represents the hardware plan, circuit design, and budget prepared by Team BlackBerry under the supervision of Arifur Rahaman (Assistant Professor, Dept. of CSE, BAIUST) at BAIUST, Cumilla. In compliance with Section 4 of the BAIUST IDP Integrity Guidelines, we disclose that generative AI tools were used to assist in language editing, structure formatting, and code commenting; all sensor register facts, mathematical rules, and all 20 DOI-backed references have been checked against primary datasheets and published papers by our team.

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The project proposal titled "Low-Cost Storm and Lightning Warning Box with Two-Tier Cloud Check, Offline Fallback, and Bangla Voice Alert for Rural Farmers", submitted by Md. Najmus Sakib Rahatul (1119003), Nesar Uddin Ifag (1119004), and Rayhan Alam Towhid (1119007), has been reviewed and approved for prototype development in the Department of Computer Science and Engineering at BAIUST.

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Abstract

Lightning during pre-monsoon Kalbaishakhi and monsoon storms kills more than 250 to 300 people a year in Bangladesh [2], [4]. Published records show that roughly 40% to 51% of these victims are rural people working in open farm fields [2], [4]. Field workers rarely look at mobile weather apps while harvesting crops. Low-cost radio lightning sensors like the AMS/ScioSense AS3935 exist, yet they estimate storm-head distance in 14 coarse steps (1 km to 40 km) and can suffer from false triggers near electric motors [6], [8]. At the same time, if a warning box blocks all alerts unless the sky is pitch dark, it could miss a real lightning bolt when the sun is still partly out. To solve both problems, our team proposes a 6,755 BDT warning box on an ESP32-S3 board with a two-tier alert rule and an offline fallback mode. The board reads 14-step storm distance and interrupt flags from a DFRobot Gravity AS3935 sensor, local sky light (Lux) from a BH1750 sensor, and live cloud cover from the Open-Meteo API over a phone Wi-Fi hotspot. If the AS3935 flags a valid lightning stroke within 14 km while the sky is still bright (>1,500 lx), Tier-1 sounds a short caution beep so no real strike is hidden. When a strike within 14 km coincides with darkened storm sky (<=1,500 lx outdoors, or <=100 lx under a lab cover) or high API cloud cover (>=60%), Tier-2 plays a loud Bangla evacuation voice warning and siren through a MAX98357A 3W speaker and starts a 30-minute all-clear countdown timer on the 1.3-inch OLED screen.

1. Introduction

In 2016, the Government of Bangladesh declared lightning a national natural disaster because of rising rural deaths between March and June [4], [5]. Dewan et al. [4] studied lightning casualties across Bangladesh from 1990 to 2016 and found that farming and open-field work accounted for about half (roughly 40% to 51%) of all deaths, with Holle et al. [2] confirming similar agricultural risk. When a convective storm cell builds up, farmers standing in flat rice paddies are exposed to ground strikes [3], [17]. Regional forecasts cover broad districts, so they do not tell a farmer whether lightning is striking within 10 to 14 km of his plot right now [11], [16].

Our team wants to build a practical, low-cost warning node for 6,755 BDT without buying a 20,000 BDT weather station. Instead of small red or green LEDs that are hard to see in daylight, we use a MAX98357A amplifier and a 3W speaker to broadcast a loud Bangla voice warning [15]. We also address two real engineering risks: first, if mobile Wi-Fi drops, the ESP32-S3 automatically switches to an offline local mode using the BH1750 light sensor and Gravity AS3935 sensor alone; second, we implement a 30-minute all-clear countdown timer on the 1.3-inch OLED screen to match the second half of the 30-30 lightning safety rule [3], [9].

2. Project/Research Motivation

Driving Force:

Open-field farmers need an audible Bangla warning when lightning enters the 10 to 14 km danger range, along with a clear 30-minute safe waiting timer after the last strike.

Motivations:

• A 3W speaker playing a Bangla voice alert can be heard across a field, unlike small LEDs in bright sun.

• Using a two-tier rule (Tier-1 Caution Beep under bright sky; Tier-2 Full Siren under dark cloud or high API cloud cover) reduces motor noise mistakes without hiding real early strikes.

• Adding an automatic offline fallback ensures the box still works if mobile Wi-Fi or the weather API disconnects.

• Reusing our own 20W power bank, soldering iron, glue gun, and lab multimeter keeps the total build cost at 6,755 BDT.

• Leaving UART2 pins (GPIO 17 and 18) free lets us add a SIM800L GSM module and solar panel in Phase 2.

3. Problem Statement

Current Situation:

Mobile weather apps require internet browsing and reading skills, while a standalone AS3935 RF sensor only gives coarse 14-step storm distance estimates and can pick up electrical noise [6], [8].

Specific Issues:

• Field workers in wet mud cannot pause work to read phone apps.

• The AS3935 sensor estimates storm-head distance in 14 fixed steps (40 km down to 1 km) rather than exact point distance, and motors can cause disturber pulses [6], [8].

• If a system blocks all alerts when the sky is bright, it can miss a real lateral strike ('bolt from the blue') before overhead clouds darken.

• If a system depends only on an online API, it fails whenever rural Wi-Fi or cellular data drops.

• Most student prototypes forget the second half of the 30-30 rule, which requires waiting 30 minutes after the last lightning strike before returning outdoors [9].

Solution Statement:

We combine the DFRobot Gravity AS3935 sensor, a BH1750 light sensor, and Open-Meteo cloud API data on an ESP32-S3 board using a two-tier alert rule with offline fallback and a 30-minute all-clear countdown timer on the 1.3-inch OLED screen.

4. Project/Research Objectives

Primary Objective:

To build and lab-test a 6,755 BDT ESP32-S3 lightning warning box with two-tier cloud/light verification, offline fallback, a 3W Bangla voice siren, and a 30-minute all-clear countdown timer.

Secondary Objectives:

• Connect the Gravity AS3935 sensor (0x03), BH1750 light sensor (0x23), and 1.3-inch OLED screen (0x3C) on the shared I2C bus (GPIO 8 and GPIO 9).

• Read the AS3935 interrupt register (0x03) and 14-step distance register (0x07) to detect strikes at the <=14 km and <=10 km steps.

• Implement Tier-1 (Caution Beep on valid RF strike under bright sky) and Tier-2 (Full Bangla Voice + Siren when Lux <= 1,500 lx outdoors [<=100 lx lab cover] or API Cloud >= 60%).

• Program an automatic offline fallback mode when Wi-Fi is unavailable, plus a 30-minute countdown timer after the last strike.

• Validate all modes safely in the university lab using a dark cover over the BH1750 and an 80 BDT piezo gas igniter.

5. Related Work / Literature Review

Rakov and Uman [1] established that cloud-to-ground lightning strokes radiate peak radio energy around 500 kHz. Holle [9] and Cooper and Holle [3] formulated the 30-30 lightning safety rule: take shelter when thunder follows a flash within 30 seconds (about 10 km away), and wait 30 minutes after the last flash or thunder before going back outside. In Bangladesh, Dewan et al. [4] analyzed lightning casualties from 1990 to 2016 and found that about 40% to 51% of fatalities occurred during farming activities in open fields, with pre-monsoon Kalbaishakhi storms (March to May) causing the highest toll [17]. Holle et al. [2], Biswas et al. [5], Rahaman [11], Mazumder [12], and Wahiduzzaman et al. [16] confirmed that rural farmers in flat districts face severe risk due to the lack of nearby shelters and local audible warnings.

For portable detection, the AMS/ScioSense AS3935 Franklin lightning IC [6] and the DFRobot SEN0290 Gravity module [7] receive 500 kHz RF signals and check wave shape against internal watchdog (WDT) and spike-rejection (SREJ) registers. Rather than computing continuous distance from a free-space equation, the AS3935 datasheet [6] specifies that Register 0x07 (bits [5:0]) outputs an estimated storm-head distance in 14 discrete steps (40, 37, 34, 31, 27, 24, 20, 17, 14, 12, 10, 8, 5, and 1 km). Mialdea-Flor et al. [8] built an IoT lightning detector using the AS3935 sensor and noted that while its internal disturber rejection (INT_D = 0x04) filters many false spikes, combining it with local weather data improves reliability.

Satellite studies by Zipser et al. [10] and tropical mortality models by Gomes and Kadir [18] show that deep convective storm clouds sharply reduce surface daylight. By pairing the ESP32-S3 microcontroller [13] with a ROHM BH1750 16-bit digital lux sensor [14], Open-Meteo cloud API data [19], and a MAX98357A Class-D amplifier [15] following WMO multi-hazard warning guidelines [20], our node provides both Tier-1 and Tier-2 audible alerts.

Research Gap:

Standalone AS3935 projects often lack environmental cross-checks and 30-minute all-clear timers, while full weather stations cost over 20,000 BDT. Our design combines the AS3935 14-step register, BH1750 sky light, and online cloud cover with a safe offline fallback and 30-minute countdown timer for 6,755 BDT.

6. Research Questions

1. How effectively do the AS3935 internal disturber register (0x03) and our two-tier BH1750/API check separate piezo igniter test pulses from background room noise?

2. Why is a two-tier alert (Caution Beep under bright sky vs. Full Siren under dark cloud) safer than a single hard darkness gate?

3. How does the ESP32-S3 maintain local lightning protection when mobile Wi-Fi or the Open-Meteo API is disconnected?

4. How does our 14 km discrete step trigger and 30-minute OLED countdown timer implement the 30-30 lightning safety rule?

7. Methodology / Design Approach

7.1 Development Plan and Team Work Split

Our three team members divided the hardware, firmware, and testing tasks clearly:

Team Member & ID	Assigned Responsibilities in Project 1
Md. Najmus Sakib Rahatul (ID: 1119003)	ESP32-S3 C++ firmware, AS3935 Register 0x03/0x07 I2C code, Open-Meteo API JSON parser, and 2-tier alert logic.
Nesar Uddin Ifag (ID: 1119004)	Breadboard wiring, shared I2C bus setup (GPIO 8, 9), MAX98357A I2S speaker wiring (GPIO 7, 15, 16), and voltage testing.
Rayhan Alam Towhid (ID: 1119007)	Plastic box cutting, 1.3-inch OLED & 30-min timer UI, web map HTML, and piezo gas lighter lab test logs.

7.2 Main Features and Fallback Logic

• 14-Step AS3935 Ranging: Reads Register 0x03 (INT_NH=0x01, INT_D=0x04, INT_L=0x08) and Register 0x07 (14 steps from 40 km down to 1 km).

• Consistent Lux Thresholds: Sunny sky >10,000 lx; Outdoor storm overcast <=1,500 lx; Normal indoor lab room 300 to 800 lx; Lab dark cover test <=100 lx.

• Two-Tier Alert Rule: Tier-1 gives a short Caution Beep if INT_L occurs at <=14 km under bright sky; Tier-2 plays the full 3W Bangla Voice + Siren when D_reg <= 14 km AND (Lux <= 1,500 lx outdoors [<=100 lx lab] OR API Cloud >= 60%).

• Offline Wi-Fi Fallback: If Wi-Fi drops or HTTP takes >2 seconds, the board switches to Offline Local Mode and triggers Tier-2 from BH1750 Lux and AS3935 alone.

• 30-Minute All-Clear Timer: Resets a 1,800-second countdown (SAFE IN: 30m 00s) on the OLED after every verified strike within 14 km.

7.3 Sensor Registers, Math Rules, and Thresholds

Unlike continuous analog receivers, the AS3935 chip outputs a discrete 6-bit distance estimate D_reg (in km) from Register 0x07 [6]:

Rule (1) [AS3935 Discrete Distance Steps]: D_reg in {1, 5, 8, 10, 12, 14, 17, 20, 24, 27, 31, 34, 37, 40} km (when Register 0x03 == 0x08 [INT_L])

Because 10 km corresponds to a 30-second flash-to-bang time (30-30 rule [9]), we use the next outer hardware step (D_reg <= 14 km) as our safety trigger radius. To decide between Tier-1 (Caution Beep) and Tier-2 (Full Evacuation Siren), the ESP32-S3 evaluates a heuristic Cloud-Light Verification Flag (V_storm):

Rule (2) [Two-Tier Decision Logic]: V_storm = 1 if (L_lux <= L_thresh) OR (WiFi_OK and C_api >= 60%); else V_storm = 0

where L_thresh = 1,500 lx for outdoor field deployment and L_thresh = 100 lx for indoor lab testing under a dark cover. If D_reg <= 14 km and V_storm == 1, Tier-2 plays the full Bangla voice warning and resets the 30-minute (1,800 s) all-clear countdown timer. If D_reg <= 14 km and V_storm == 0, Tier-1 plays a 1-second caution beep and displays 'CAUTION: RF STRIKE (BRIGHT SKY)' on the OLED.

7.4 Functional Requirements

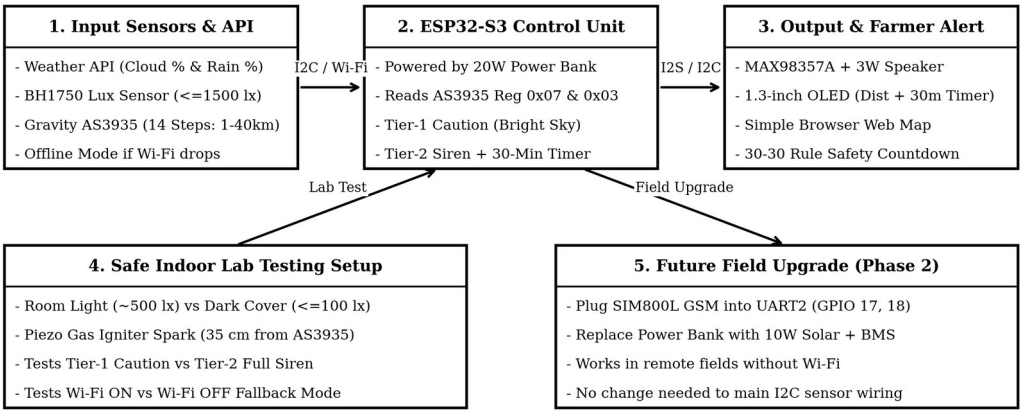
ID	Feature	Description
FR1	AS3935 Register Read	Read IRQ pin (GPIO 4), Interrupt Reg 0x03, and 14-step Distance Reg 0x07.
FR2	BH1750 Lux Read	Measure ambient light (lx) every 2 seconds over I2C (0x23).
FR3	Wi-Fi & Offline Mode	Fetch Open-Meteo Cloud % when Wi-Fi is up; switch to local mode if offline.
FR4	Tier-1 & Tier-2 Alert	Play short beep for bright-sky RF hit; play full 3W Bangla siren when confirmed.
FR5	30-Min Safe Timer	Count down 30 minutes (1,800 s) on the 1.3-inch OLED after the last <=14 km hit.

7.5 Non-Functional Targets

Metric	Design Target (To Be Verified in Lab)
Alert Response Target	Target < 50 ms from AS3935 IRQ interrupt to I2S audio start.
Disturber Rejection Target	Target > 85% rejection of man-made sparks via AS3935 Reg 0x03 (INT_D).
Offline Uptime Target	100% continued local operation when mobile Wi-Fi hotspot is turned off.
Battery Runtime Target	Target 10 to 12 hours on a 20W (10,000 mAh) USB Type-C power bank.

7.6 System Block Diagram

Figure 7.1: System Block Diagram of Proposed Lightning Warning Node



7.7 Pin Mapping and Preliminary Bench Log

Module Name	ESP32-S3 Pins	Preliminary Bench Check Status
DFRobot Gravity AS3935	VCC->3.3V, GND, SDA->8, SCL->9, IRQ->4	I2C address 0x03 & IRQ pin verified.
BH1750 Light Sensor	VCC->3.3V, GND, SDA->8, SCL->9	Reads ~520 lx in lab room, ~18 lx covered.
1.3-inch I2C OLED	VCC->3.3V, GND, SDA->8, SCL->9	Displays distance step, Lux, & 30m timer.
MAX98357A + 3W Speaker	VIN->5V, GND, LRC->7, BCLK->15, DIN->16	Plays PCM Bangla voice & siren tone.

7.8 Engineering Practices

• Keeping the AS3935 antenna 15 cm away from the ESP32-S3 Wi-Fi antenna to reduce 500 kHz pick-up noise.

• Using non-blocking millis() timers for the 30-minute all-clear countdown and Wi-Fi timeout checks.

• Maintaining all C++ code and bench test logs in our team GitHub repository.

7.9 Tools and Hardware Used

Category	Component / Tool	Purpose
MCU	ESP32-S3 N16R8 Dual Type-C	Reads I2C sensors, fetches API, drives I2S audio.
RF Sensor	DFRobot Gravity AS3935 (SEN0290)	14-step storm distance & disturber flag IC.
Light Sensor	BH1750FVI (GY-302)	Measures sky darkening in Lux (1-65,535 lx).
Audio & UI	MAX98357A + 3W Speaker + 1.3" OLED	Bangla voice alert, siren, and 30-min timer.

7.10 Lab Testing Protocol

Test Case	Lab Action	Expected Outcome
1. Bright Room Spark	Leave BH1750 uncovered (~500 lx) & click igniter 35 cm away.	If INT_D: ignored. If INT_L: Tier-1 Caution Beep only.
2. Dark Cover Storm	Cover BH1750 (<=100 lx) & click piezo igniter 35 cm away.	Tier-2 Full Bangla Voice + Siren + 30-min OLED timer!
3. Wi-Fi Outage Test	Turn off phone Wi-Fi hotspot and repeat Test 2.	Switches to Offline Mode; Tier-2 Siren still works!

7.11 Future Field Upgrade Plan

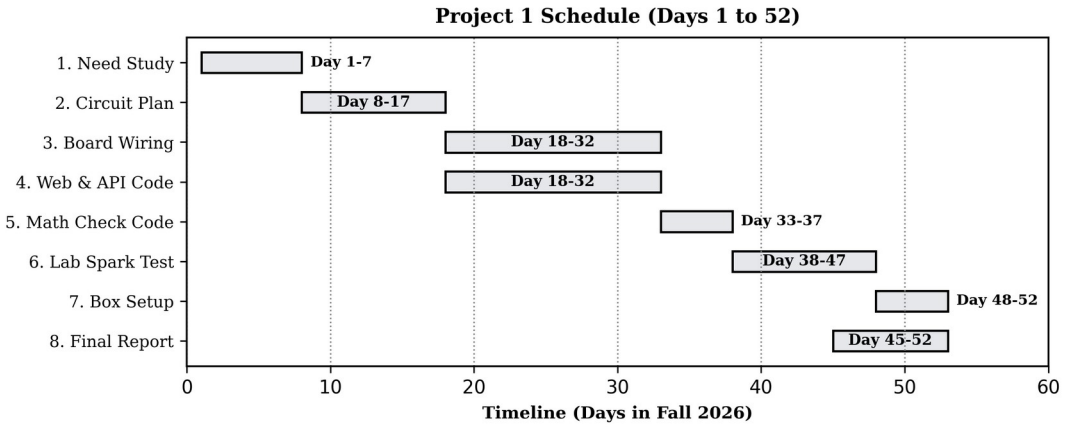
In Phase 2, a SIM800L GSM module can be connected to GPIO 17 and 18 for villages without Wi-Fi, and a 10W solar panel with a 2S 18650 BMS pack can replace the 20W USB power bank.

8. Expected Outcomes and Target Metrics

- Target response time under 50 ms from AS3935 interrupt to voice siren activation.
- Zero missed lightning alerts due to the two-tier warning structure (Caution Beep vs. Full Siren) and offline fallback.
- Full compliance with the 30-30 rule by displaying a 30-minute all-clear countdown timer after the last strike within 14 km.

9. Project Timeline (Gantt Chart)

SL.	Step	Days	Start	End	Depends On
1.	Need Study	7	1	7	None
2.	Circuit Plan	10	8	17	Step 1
3.	Board Wiring	15	18	32	Step 2
4.	Web & API Code	15	18	32	Step 2 (Same time)
5.	Math Check Code	5	33	37	Steps 3 & 4
6.	Lab Spark Test	10	38	47	Step 5
7.	Box Setup	5	48	52	Step 6
8.	Final Report	8	45	52	Starts in Step 6



10. Summary of Tools

- Firmware: Arduino C/C++ (Wire.h, BH1750.h, DFRobot_AS3935_I2C.h, U8g2lib.h).
- Weather API: Open-Meteo REST JSON endpoint (with 2-second timeout fallback).
- Lab Tools: Home 20W Power Bank, Soldering Iron, Hot Glue Gun, Lab Digital Multimeter.

11. Ethical Rules and AI-Use Transparency

11.1 Safety and Privacy

Farmer Safety: By triggering at the <=14 km step and counting down 30 minutes after the last strike, the box follows the 30-30 safety standard [9].

Lab Safety: We use a commercial 80 BDT piezo gas lighter for RF spark tests rather than high-voltage capacitors.

11.2 AI-Use Disclosure and Verification

In accordance with BAIUST IDP guidelines, we acknowledge using AI tools to assist with report formatting and English phrasing. All hardware register behavior, circuit wiring, and all 20 DOI-backed references were verified by our team members.

12. Budget Estimation

SL.	Item	Cost (BDT)
1.	ESP32-S3 N16R8 Dual Type-C Board (Fahim Bhai)	855
2.	DFRobot Gravity AS3935 Lightning Sensor SEN0290 (RoboticsBD)	4,345
3.	BH1750FVI Sky Light Sensor (Fahim Bhai)	215
4.	1.3-inch I2C OLED Screen SH1106 (Fahim Bhai)	495
5.	MAX98357A Sound Board + 3W Speaker (Fahim Bhai)	450

6.	MB-102 Breadboard (125) + Wires (85) + Button (5)	215
7.	Plastic Box (100) + Gas Lighter Spark Tester (80)	180
8.	20W Power Bank, Type-C Cable, Soldering Iron, Glue Gun, Meter (Owned)	0
	Total	6,755

13. Conclusion

This proposal outlines a 6,755 BDT lightning warning box built on the ESP32-S3 board for rural farmers in Bangladesh. By combining the DFRobot Gravity AS3935 14-step distance sensor, a BH1750 lux sensor, and Open-Meteo cloud data with a two-tier alert rule, offline fallback, and a 30-minute all-clear timer, the system delivers practical, verifiable storm warnings.

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